```
# IMPORTANT: RUN THIS CELL IN ORDER TO IMPORT YOUR KAGGLE DATA SOURCES,
# THEN FEEL FREE TO DELETE THIS CELL.
# NOTE: THIS NOTEBOOK ENVIRONMENT DIFFERS FROM KAGGLE'S PYTHON
# ENVIRONMENT SO THERE MAY BE MISSING LIBRARIES USED BY YOUR
# NOTEBOOK.
import kagglehub
osmi_mental_health_in_tech_survey_path = kagglehub.dataset_download('osmi/mental-health-in-tech-survey')

print('Data source import complete.')
```

```
# This Python 3 environment comes with many helpful analytics libraries installed
# It is defined by the kaggle/python Docker image: https://github.com/kaggle/docker-python
# For example, here's several helpful packages to load

import numpy as np # linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)

# Input data files are available in the read-only "../input/" directory
# For example, running this (by clicking run or pressing Shift+Enter) will list all files under the input directory

import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))

# You can write up to 20GB to the current directory (/kaggle/working/) that gets preserved as output when you create a version using "Save & Run All"
# You can also write temporary files to /kaggle/temp/, but they won't be saved outside of the current session
```

```
/kaggle/input/datasets/organizations/osmi/mental-health-in-tech-survey/survey.csv
```

Mental Health in Tech: EDA + Machine Learning

Goal: Predict whether a person seeks mental health treatment based on workplace and personal factors.

```
df = pd.read_csv("/kaggle/input/datasets/organizations/osmi/mental-health-in-tech-survey/survey.csv")
df.head()
```

	Timestamp	Age	Gender	Country	state	self_employed	family_history	treatment	work_interfere	no_employees	...	leave	mental_health_consequence	phys_health_
0	2014-08-27 11:29:31	37	Female	United States	IL	NaN	No	Yes	Often	6-25	...	Somewhat easy	No	
1	2014-08-27 11:29:37	44	M	United States	IN	NaN	No	No	Rarely	More than 1000	...	Don't know	Maybe	
2	2014-08-27 11:29:44	32	Male	Canada	NaN	NaN	No	No	Rarely	6-25	...	Somewhat difficult	No	
3	2014-08-27 11:29:46	31	Male	United Kingdom	NaN	NaN	Yes	Yes	Often	26-100	...	Somewhat difficult	Yes	
4	2014-08-27 11:30:22	31	Male	United States	TX	NaN	No	No	Never	100-500	...	Don't know	No	

5 rows × 27 columns

```
# importing libraries
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
```

```
# Handle missing values
df = df.dropna()

# Fix gender
# Remove outliers (age)
df = df[(df['Age'] >= 15) & (df['Age'] <= 70)]
```

EDA Summary

Key Insights from EDA

- Family history strongly affects treatment
- Work interference increases likelihood
- Benefits improve help-seeking behavior

```
# Convert categorical to numeric
le = LabelEncoder()
```

```
for col in df.columns:
    if df[col].dtype == 'object':
        df[col] = le.fit_transform(df[col])
```

```
# Define X and y
x = df.drop("treatment", axis=1)
y = df["treatment"]
```

```
x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.2, random_state=42
)
```

```
model = LogisticRegression(max_iter=1000)
model
```

```
LogisticRegression
LogisticRegression(max_iter=1000)
```

```
model.fit(x_train, y_train)
```

```
LogisticRegression
LogisticRegression(max_iter=1000)
```

```
y_pred = model.predict(x_test)
y_pred
```

```
array([1, 1, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1])
```

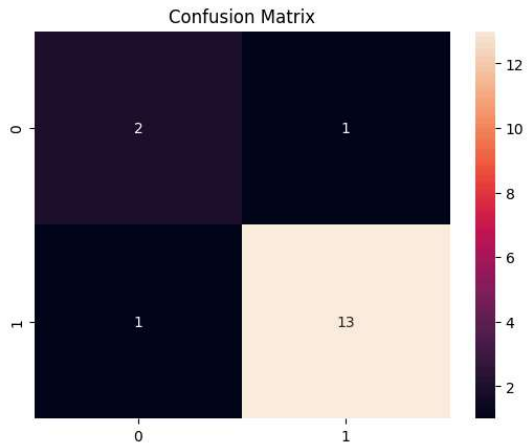
```
print("Accuracy:", accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

```
Accuracy: 0.8823529411764706
              precision    recall  f1-score   support

           0       0.67      0.67      0.67         3
           1       0.93      0.93      0.93        14

    accuracy                           0.88        17
   macro avg       0.80      0.80      0.80        17
weighted avg       0.88      0.88      0.88        17
```

```
sns.heatmap(confusion_matrix(y_test, y_pred), annot=True, fmt='d')
plt.title("Confusion Matrix")
plt.show()
```




```
<Figure size 640x480 with 0 Axes>
```

```
plt.savefig("confusion_matrix.png")
```

```
<Figure size 640x480 with 0 Axes>
```

Conclusion

- Logistic Regression successfully predicts treatment likelihood
- Family history and work interference are key factors
- Companies can use such models to identify risk and improve support systems